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SUMMARY

Generative AI Engineer with experience building and shipping AI-powered systems and products. I work mainly with Python, FastAPI, and TypeScript, with a focus on agentic workflows, RAG, automation, and backend systems. My experience includes building production systems for incident automation, developing full-stack AI applications, and creating open-source developer tools.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, SQL

AI Frameworks & Tools: AutoGen, LangGraph, LangChain, Google ADK, AI SDK, Qdrant, Pinecone, Prompt Engineering, MCP, RAG, LLMs, Claude Code, OpenAI Codex

Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, MSSQL, Redis, MongoDB, SQLite, Node.js, Express

Frontend & Platform: Next.js, React, TanStack Start, Effect TS, AWS Lambda, Google Cloud, Docker, Kubernetes, Cloudflare, Grafana Loki, GitLab CI/CD, CI/CD

WORK EXPERIENCE

Tata Consultancy Services

Nov 2024 – Present

AI Engineer

Chennai, TN

- Built Python-based production incident automation that integrates **Grafana Loki** anomaly signals with ServiceNow REST APIs, cutting operational MTTR by 63%.
- Reduced average incident resolution time from 25-30 minutes to 6-8 minutes by integrating a **Qdrant-backed RAG** context-engineering process into a multi-agent incident-resolution workflow.
- Led migration of a monolithic Next.js frontend (50+ modules) to a micro-frontend architecture, halving build time and raising deployment throughput 2.5x.
- Partnered with the Google Cloud team to ship a platform representing TCS at **Google Cloud Next 2026**, showcasing 120+ AI offerings to Google and TCS sales teams.
- Owned CI/CD across infrastructure, sales, and business-analysis teams, supporting 25+ production deploys/month with zero critical rollbacks.

PROJECTS

PROFESSIONAL

Agentic AMS Platform

Feb 2025 – Sep 2026

Python, AutoGen, FastAPI, MCP, Qdrant, Grafana Loki, ServiceNow

[Case Study](#)

- Designed a **FastAPI and Microsoft AutoGen** multi-agent system that detects, diagnoses, and remediates production failures through MCP-enabled tool calling.
- Built an observability agent that ingests **Grafana Loki** alerts over WebSockets, deduplicates ServiceNow tickets, and runs document ingestion, chunking, embeddings, and semantic retrieval through a Qdrant RAG index.
- Implemented a **Planning - Executor - Validator** agent workflow with human-in-the-loop approval for infrastructure rollbacks, structured validation, and automated pod health checks.

Google GTM Portal

Jan 2026 – Apr 2026

Python, Next.js, FastAPI, GCS, Google Cloud

[Case Study](#)

- Architected a typed FastAPI and Google Cloud Storage catalog backend with a Next.js frontend, delivering sub-second loads and a 96% Lighthouse score.
- Implemented **hybrid keyword and vector retrieval** to improve discovery across a catalog of 120+ AI offerings.

OPEN SOURCE

Cortex

Apr 2026 – May 2026

Effect TS, TypeScript, Pinecone, Qdrant

[Docs](#)

- Authored an Effect-native **vector database ORM** with structured concurrency, typed dependency injection, and compositional error handling.
- Shipped provider adapters (Zvec, PageIndex) so multi-tenant RAG pipelines stay portable across vector backends.
- Raised batch insert throughput to 3.2x baseline using non-blocking batch stream pipelines on Pinecone and Qdrant.

Ascendant

Sep 2025 – Present

Python, TypeScript, Effect, Agent Skills

[Docs](#)

- Built local-first astrology SDKs: **Effect TS** (astro-ascendant, 600/week peak) with composable **Services/Layers**, typed **Dependency injections**, typed errors, and concurrent pipelines, plus Python on PyPI covering 16 charts, Dasha, 360-cell Ashtakavarga, and transits.
- Shipped **Agent Skill** (18 installs via skills.sh) with init/setup/analysis for 6+ hosts and 5-module calculator for KP-first grounded readings from computed charts.

Retail Platform

Jul 2024 – Jun 2026

Go, Next.js, AWS Lambda, Cloudflare D1, Workers

[GitHub](#)

- Designed 7 event-driven **Go microservices** on AWS Lambda (search, recommendations, notifications, orders) behind a Next.js admin/CMS for 200+ products.
- Used Cloudflare Edge caching, D1 databases, and Workers to absorb **10k req/min spikes**, cutting **p99 API latency from 450ms to 85ms**.

ACHIEVEMENTS

- OSS Contributor to Vercel (ai-elements), OpenCode, and mattpocock/skills.
- Open Source Impact: astro-ascendant reached **600 npm downloads** with a mirrored Python package on PyPI.
- TCS Special Recognition Award (2025) for standout performance in the company-wide AI Ignite Hackathon.

EDUCATION

Amrita Vishwa Vidyapeetham

Sep 2020 – May 2024

B.Tech, Computer Science and Engineering

[Paper](#)

- Research: Complex or Irregular Lung Nodule Detection from Computed Tomography Scans.